

Dairy Farm Studio • Enterprise DBMS Management System

Presentation-III: Full-Stack Working Demonstration & UI Implementation

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Database Engine Live MySQL 8.0 Community Server	Presentation Part III of Series — Working Demo & Architecture

8

Studio Modules

Dashboard, Herd Registry, Milk Yields, Quality Lab, Feed Stock, Billing, SQL Studio, Schema Inspector

14

Base Relations

Active tables in dairy_farm_db with 1 summary view and 4 active triggers

1.38ms

Query Latency

Sub-2 millisecond telemetry executed natively against MySQL 8.0.46 on localhost:3306

Working Demonstration & Architecture Overview

Live Full-Stack Architecture Interfacing Directly with MySQL 8.0

8 Studio Modules

Comprehensive workbench covering **Dashboard KPIs**, **Herd Registry**, **Milk Yields**, **Quality Lab**, **Feed Stock**, **3NF Billing**, **SQL Studio**, and **Schema Inspector** — each module independently functional and tab-navigable within the single-page application.

Obsidian & Emerald UI

Modern dark studio aesthetic inspired by **Supabase Studio** and **Linear.app** from VoltAgent/awesome-design-md. Deep obsidian canvas with hairline panels, Supabase emerald accents, and Linear lavender navigation — zero heavy CSS framework dependencies.

1.38 ms Telemetry

Sub-2 millisecond query latency executed natively against **MySQL Community Server 8.0.46** on localhost:3306. Live performance metrics surfaced directly in the dashboard KPI strip on every page load and refresh cycle.

Before/After Audit Protocol

Real-time slide-down audit drawer displaying **database row count deltas** on every INSERT and DELETE operation. Provides immediate visual confirmation of transactional integrity and trigger execution across all 14 base relations.

3-Tier System Architecture & Tech Stack

Enterprise Decoupled Stack Connected to MySQL 8.0



Presentation Layer

HTML5 + Vanilla CSS3, Modular SPA, 8 modules



API Gateway

FastAPI (Python 3.12), PyMySQL, Pydantic



Database Engine

MySQL 8.0, dairy_farm_db, 14 relations

Tier 1: Presentation Layer

- **HTML5 + Vanilla CSS3:** High-performance semantic frontend with zero heavy CSS framework dependencies.
- **Modern Studio System:** Deep obsidian canvas (#090a0d), hairline panels (#12141a), Supabase emerald (#3ecf8e), and Linear lavender (#5e6ad2).
- **Modular SPA:** Tabbed navigation switching seamlessly between 8 functional modules.

Tier 2: Application API Gateway

- **FastAPI (Python 3.12):** Asynchronous REST backend handling structured API endpoints.
- **PyMySQL Driver:** Native database driver with parameterized SQL execution and connection lifecycle handling.
- **Pydantic Validation:** Strong request validation enforcing types, formats, and non-empty payloads.

Tier 3: Database Engine

- **MySQL 8.0 Community:** Active database dairy_farm_db with **14 base relations**, **1 summary view**, and **4 active triggers**.

Live Executive Telemetry & KPI Dashboard

Real-Time Aggregated Farm Production and Commercial Overview

50

Active Cattle

Livestock master records tracked in table animal

₹18,905

Sales Revenue

3NF reconciled commercial revenue computed via trigger after_sale_item

991.0L

Milk Harvested

Cumulative litres across 50 production sessions in milk_yield

0

Stock Alerts

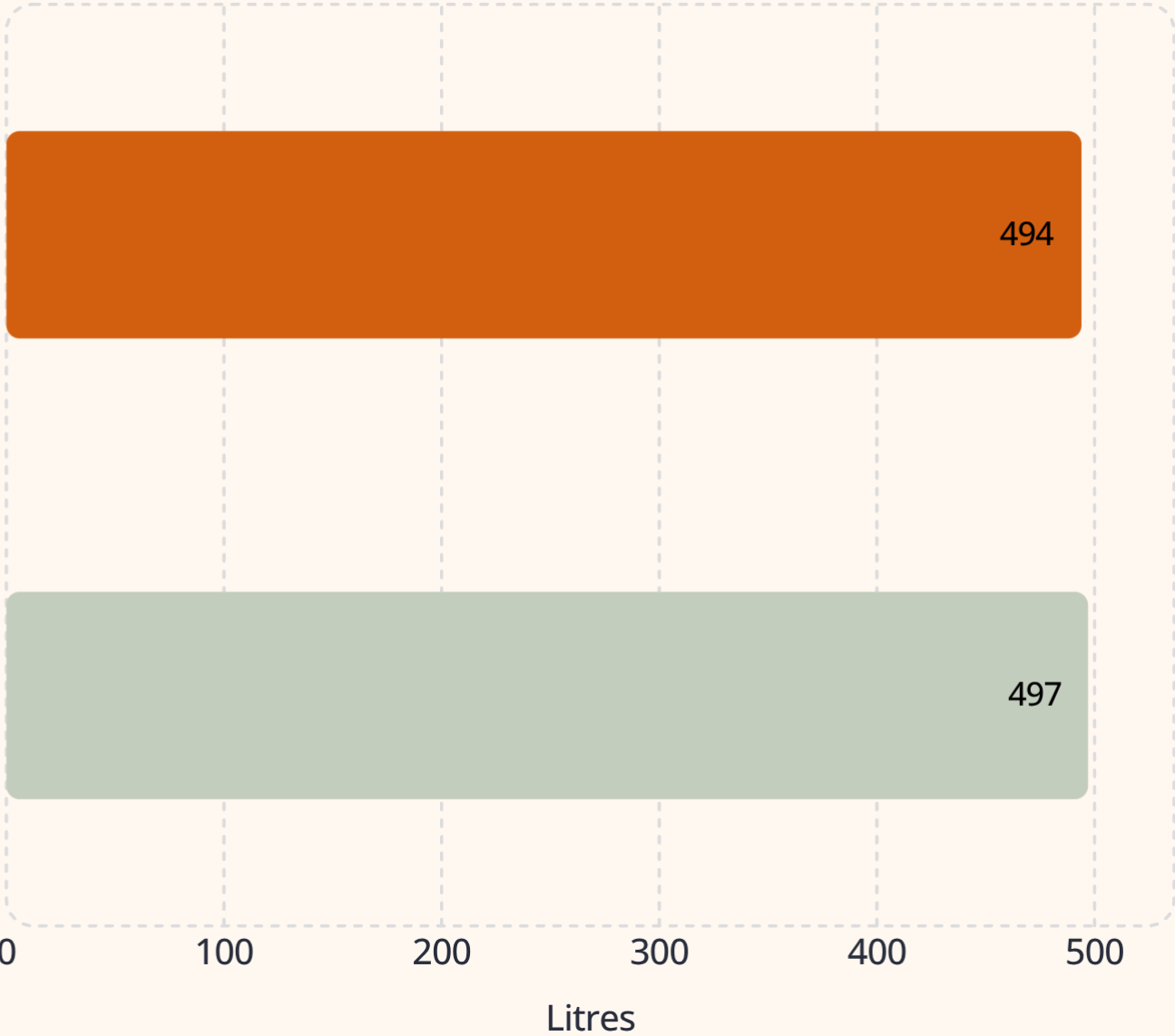
Feed commodity levels safe above reorder thresholds in feed_item

Harvest Distribution

Milking Session

Morning Milking

Evening Milking



Morning and evening sessions are near-perfectly balanced at **494.0 L** and **497.0 L** respectively, totalling **991.0 litres** across 50 recorded production sessions.

Laboratory Quality Averages

3.95% Butterfat

Average fat percentage across all quality test records in quality_test.

8.60% Solids-Not-Fat (SNF)

Average SNF percentage indicating milk solids composition quality.

1.030 Specific Density

Average specific gravity confirming milk density within acceptable commercial standards.

Cattle Master Registry & CRUD Management

Biological Lifecycle Tracking with Foreign Key Cascades



Authentic Cattle Identities

Every animal displays a registered name (e.g. **Nandini**, **Kamdhenu**, **Gauri**, **Mangala**) alongside official ear tags **A1001 – A1050**.



Dynamic Search & Filtering

Real-time client-side search across cattle **names**, **ear tags**, and **breed classifications** with instant results on every keystroke.



Operational Status Filtering

Instant filter by **Active**, **Sold**, **Deceased**, or **Transferred** status across the full livestock registry.



Cascade Delete Safety

Safety confirmation dialog detailing child record purges in `health_visit`, `vaccination`, and `milking_session` before any DELETE is committed.

Real Commercial & Feed Entities

Warehouse commodities replace placeholder numbers with authentic identifiers:

- **Feed Commodities:** Alfalfa Hay, Corn Silage, Cottonseed Cake
- **Cooperative Buyers:** Amul, Mother Dairy, Nandini, Heritage Fresh

Atomic Modal Registration

Form validation enforces the UNIQUE KEY constraint on cattle ear tags at the application layer via **Pydantic** before any INSERT reaches the MySQL engine — preventing duplicate registrations and ensuring referential integrity across all downstream relations.

Interactive SQL Studio: In-Browser Query Engine

Sub-2ms Live Query Execution with Dynamic Tabular Results

Live Query: Full Animal Production Profile

The following query demonstrates an **8-table JOIN** traversing the complete production chain — from animal identity through milking session, yield measurement, quality testing, sale item, sale header, and cooperative buyer — executed live against dairy_farm_db in under 2 milliseconds:

```
SELECT
  a.animal_id,
  a.animal_tag,
  b.breed_name,
  my.quantity_litres,
  qt.fat_percentage,
  qt.snf_percentage,
  si.quantity_sold,
  s.total_amount,
  bu.buyer_name
FROM animal a
JOIN breed b
  ON a.breed_id = b.breed_id
JOIN milking_session ms
  ON a.animal_id = ms.animal_id
JOIN milk_yield my
  ON ms.session_id = my.session_id
LEFT JOIN quality_test qt
  ON my.yield_id = qt.yield_id
LEFT JOIN sale_item si
  ON my.yield_id = si.yield_id
LEFT JOIN sale s
  ON si.sale_id = s.sale_id
LEFT JOIN buyer bu
  ON s.buyer_id = bu.buyer_id
WHERE a.animal_id = 20;
```

Query Architecture Breakdown

01

Identity Resolution

animal → breed via breed_id FK. Resolves ear tag and breed classification for the target animal.

02

Production Chain

milking_session → milk_yield via session_id FK. Retrieves raw quantity in litres per harvest event.

03

Quality Layer

LEFT JOIN quality_test on yield_id. Returns fat%, SNF% — nullable where lab test not yet recorded.

04

Commercial Resolution

LEFT JOIN sale_item → sale → buyer. Resolves quantity sold, total amount, and cooperative buyer name via 3NF billing schema.

 All JOINs are parameterized via PyMySQL driver. The WHERE a.animal_id = 20 predicate is injected as a bound parameter — never interpolated — preventing SQL injection at the API gateway layer.